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UNIT-5

Communication Engineering

Definitions :-

(1) BANDWIDTH :- It is the range of frequencies over which information signal is transmitted.

(2) Channel Bandwidth :- It is defined as the range of frequencies which a channel can pass through without any distortion.

(3) Frequency spectrum :- It is the representation of a signal in freq. domain. It Consists of amplitude and phase spectrum of signal.

(4) Base Band Signal :- It is defined as the message signal in the original form. Message signal may be in the form of text, sound, picture etc. Base band signal is also called as Modulating Signal.

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Modulation :- In modulation, some parameters of the carrier wave such as Amplitude, Frequency or phase is varied in accordance with the modulating signal, is called as Modulation.

Modulation is done at the transmitter side, and the resultant signal is called as Modulated Signal.

→ Modulated signal is also called as Broad Band Signal.

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IEEE Frequency spectrum | EM spectrum and its Applications :- (RADIO FREQUENCY SPECTRUM) :-

Frequency Band	Applications
1. 30HZ - 300HZ Extremely Low Freq. (ELF)	In power Transmission
2. 300HZ — 3 KHZ Voice Frequencies (VF)	Audio Applications
3. 3 KHZ — 30 KHZ Very Low Frequencies (VLF)	Navy, Military Applications
4. 30 KHZ — 300 KHZ Low Frequencies (LF)	Aeronautical and Marine Navigation
5. 300 KHZ - 3 MHZ Medium Frequencies (MF)	AM Radio BroadCast
6. 3 MHZ - 30 MHZ High Frequencies (HF)	short Wave transmission
7. 30 MHZ - 300 MHZ Very High Frequencies (VHF)	TV BroadCasting
8. 300 MHZ — 3 GHZ Ultra High Frequencies (UHF)	Cellular phones, Military Applications.

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9.	3 GHz - 30 GHz Super High Frequencies (SHF)	Radar And Satellite Communication
10.	30 GHz - 300 GHz Extreme High Frequencies (EHF)	Satellite and specialized Radars.

M. Imp

Need for Modulation :- Modulation is Needed due to Following reasons:-

(1) **To Reduce the height of Anteena :-** For the transmission of radio signals, the Anteena height must be a multiple of $\frac{\lambda}{4}$.

→ Minimum Height of Anteena For $\boxed{F = 10\ KHz}$ is -

$$\frac{\lambda}{4} = \frac{c}{f \times 4} = \frac{3 \times 10^8}{10 \times 10^3\ Hz \times 4} = 75m$$

→ Therefore we can see, As the frequency increases, the height of anteena decreases.

→ Minimum Height of Anteena For $\boxed{F = 1\ MHZ}$ -

$$\frac{\lambda}{4} = \frac{c}{f \times 4} = \frac{3 \times 10^8}{1 \times 10^6 \times 4} = \underline{75m}$$

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② **Modulation Avoids mixing of Signals :-** IF baseband signals are transmitted by more than 1 transmitter then all the signals will get mixed up due to same band width (0-20KHz) and Receiver can't separate them from each other.

IF each baseband signal is used to modulate a diffrent carrier, they won't get mixed up.

③ **Increases the range of Communication :-**

The Frequency of baseband Signal is low and low Frequency signals can't travel a long distance. They get heavily attenuated.

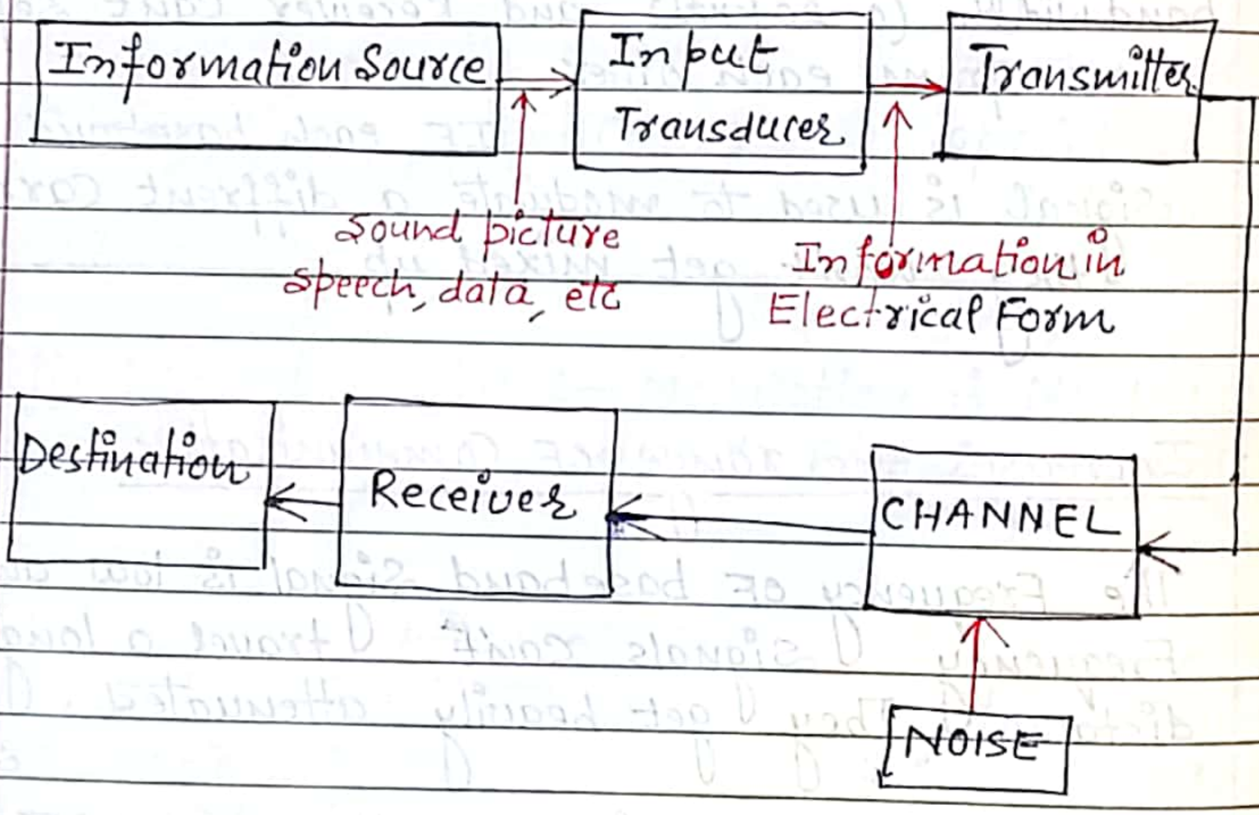
So by modulation we can increase the range of communication.

④ **Makes Multiplexing possible :-** Multiplexing is a process in which two or more signals can be transmitted over the same communication channel.

⑤ **Improves Quality Reception :-** With FM and digital communication techniques, the effect of noise is reduced to a great extent.

M.Imp

Block Diagram of Communication System :-



(1) Information Source :- Information source contains message. This message may be in the form of sound picture, speech, data, text etc.

This information may be or may not be in electrical nature.

(2) Input Transducer :- Transducer is a device which transforms one form of energy into another form.

The input

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transducer is used to Convert the information or message, which is in non-electrical form, into a time varying electrical form.

③ Transmitter :- The transmitter processes the incoming information or message signal so as to make it suitable for transmission.
Modulation is done at the transmitter.

④ CHANNEL :- channel is a medium through which the message travels from the transmitter to the receiver.
In other words the Function of channel is to provide a physical connection between the transmitter and the receiver.

⑤ Noise :- Noise may interfere with signal at any point in the communication system.
If noise is present in the channel or at the input of receiver, it harms a lot.

⑥ Receiver :- The Function of receiver is to receive the signal (modulated) and to extract the message, which is in electrical form, from the carrier signal that means De-Modulation is done at the receiver.

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⑥ Destination :- This is the Final stage which is used to Convert an electrical message signal ~~its~~ into its original Form.

For this Inverse transducer is used.